TolTEC Raw-Timestream Conditioning

Learn–Resolve–Apply Filtering, Response, and Scientific Consequences

Science-Team Rationale

Scientific contract version	v0.1
Science-team rationale revision	r0.12
Date	2026-08-21
Authority	Shared normative core in `src/common/`
Status	Scientific authority frozen by owner; implementation conformity not assessed under this contract.
Method	Implementation-blind scientific authorship

Scope of this revision. It defines canonical pair-level learning, cause-preserving hard-action union, the identical ordinary x/r operator, coordinate-specific affine corrections, the x-only donor exception, how level shifts and leakage are represented, how astronomical modes are changed, and which claims remain available afterward. It does not claim implementation conformity, representation fidelity, observational performance, science qualification, validation, or production readiness.

Executive abstract

Raw-timestream conditioning is a scientific method over exact paired raw readout coordinates, not merely a sequence of array operations. Upstream authority maps measured IQ to one x/r pair per detector occurrence. RTC retains both: x is the required science-conditioning output path, raw r remains an immutable diagnostic parent and equal contamination sensor, and requested conditioned r is an optional canonical companion on exactly the conditioned-x grid. Accepted hard evidence from x, r, or their joint behavior can alter one pair-level plan while preserving its origin. Spike-aware level-shift segmentation, later within-segment donor replacement, notches, and broad-band filters alter values, response, support, and covariance. Decimation changes temporal identity and can alias rejected modes into the retained band. Each choice can therefore change a point-source peak, effective PSF, centroid, extended-source response, calibration transfer, or scientific eligibility without changing a unit label.

SCI-RTC v0.1/r0.12 organizes those choices as **learn–resolve–apply**. Learn retains coordinate-specific and genuinely joint evidence, including source protection, direct/inferred state, support, quality, and uncertainty. Resolve combines that evidence with fixed class policy, unions only accepted hard pair-action support, and admits one complete immutable RTC pair plan. Apply executes exactly that plan and does not relearn or retune. An online adaptive filter is a different, time-varying estimator and remains unavailable unless separately authorized. Missing evidence or tolerances are not passes.

Each resolved response is evaluated against the astronomical temporal spectrum induced by the realized scan velocity, projected beam, source class, cadence, and preceding response. A fixed cutoff is not equally safe for all scan speeds or directions. A notch that removes an interference line also removes whatever astronomical response lies in its width. A detector-static `flxscale` cannot restore removed temporal modes, and calibration transfer across different RTC plans requires explicit response accounting or observational evidence.

The output is an atomic bundle of conditioned x, requested conditioned r or its exact availability state, the immutable paired raw r parent, exact mapping and plan identity, identical ordinary response on both coordinates, timing convention, support, influence, validity, uncertainty status, leakage/source-protection facts, and provenance. Raw x is not itself Stokes I and r is neither calibrated nor substituted for x. Ordinary shared stages use one common operator on both coordinates without mixing. Coordinate-specific affine offsets may differ; the existing x donor repair remains an explicit exception that invents no r and makes conditioned r unavailable over its full causal influence. Absolute `flxscale` and target atmosphere belong to a distinct downstream SCI-CAL product applied only to x. When a required predicate is unresolved, the affected operation or claim is explicitly unavailable. The shared formal core rendered in the companion Engineering Conformance PDF is the sole normative authority; Sections 1–12 explain its scientific meaning.

Reading rule. Narrative text explains the shared authority but cannot weaken a requirement, choose a missing numerical value, or convert an unavailable state into a pass.

Contents

1 Executive summary and present status

RTC is the scientific transformation from an admitted aligned paired x/r detector stream to a required conditioned x output and, when requested, its conditioned-r companion with immutable raw r parentage and explicit causal diagnostics. Its temporal, detector-mixing, selection, and cross-coordinate selection and uncertainty dependence are explicit. Its fixed-state numerical response has neither an $r \to x$ nor an $x \to r$ branch. Its purpose is not to make every sample finite or every spectrum smooth. Its purpose is to address declared contamination and sampling problems while preserving a quantitatively adequate response for the intended astronomical estimator.

The v0.1 scientific method is learned-then-fixed. Evidence may be learned from a declared population, but application begins only after resolve has produced one complete immutable plan. Online adaptation is a different estimator and is unavailable in this revision. Fixed plans use the same resolve and apply discipline even when no parameter is learned from the observation.

Every operation is interpreted through ten questions: purpose, signal model, operator, design parameters, learning, resolution and application, astronomical consequences, calibration consequences, validation, and unavailable states. Those questions are applied below to despiking, donors, masks, notches, broad-band and general filters, state, boundaries, non-finite handling, level shifts, optical leakage diagnostics, atmospheric-template evidence, and decimation.

The current document is the frozen v0.1/r0.12 scientific authority, promoted without scientific change from the verified candidate based on the approved v0.1/r0.11 architecture at comparison commit `85e1e6c6865f74f1a97e99fab465714f43877c3d`. It establishes no production numbers. The r0.10 conditioned-r decisions, r0.11 canonical-pair decisions, and seven r0.12 consistency decisions are resolved; every other open ledger decision remains open and its dependent operation unavailable. The formal core defines algebraic and semantic authority only. Implementation conformity, representation fidelity, observational performance, science qualification, and production readiness are explicitly unassessed.

> **Frozen decision boundary.** R0.12 preserves the approved r0.11 pair-level architecture while making its mapping sequence, original-pair refinement, support/availability partition, unavailable-correction influence, conditional product language, decision enumeration, and raw-r lineage explicit. R0.11 reopens pair-level evidence, selection, and action support: accepted r-origin or joint hard evidence may change conditioned-x masks, segmentation, guards, resets, support, and plan selection. It preserves the r0.10 fixed-stage filter definitions, donor-factor direction, CAL order, phase-zero identity, representative occurrences, and learn–resolve–apply lifecycle. Enumerated numerical policies remain unavailable until explicit authority resolves them.

Approved r0.11 architectural decisions retained by r0.12.

- `OWNER-090`: the canonical ordinary paired response is exactly $I_2 \otimes L_\Pi$, with identical ordinary operators and zero cross-coordinate numerical branches.
- `OWNER-091`: Learn retains coordinate-specific and genuinely joint evidence with exact origin and cause.
- `OWNER-092`: Resolve forms a cause-preserving union in one immutable pair plan; Apply neither discovers evidence nor mutates the plan.
- `OWNER-093`: accepted hard evidence pair-flags support while preserving direct, joint, and inferred cause distinctions.
- `OWNER-094`: raw validity and evidence, accepted pair action, and conditioned modification, availability, and response remain distinct layers.
- `OWNER-095`: r is an equal contamination sensor; expected optical response is not automatic pathology, and protection participates in Resolve.
- `OWNER-096`: canonical action is symmetric after separately admitted affine corrections, with bounded level-shift, notch, and donor exceptions and downstream CAL, PTC, and SCI-VAL ownership preserved.

RTC application context, resolved plan, and realized record. RTC does not partition operation availability by historical product role. Beammap, Pointing, OOF, Science, and diagnostic labels may describe requested use and consumers, but all v0.1 operation classes remain available until evidence or explicit policy disqualifies one for a particular application. Available does not mean enabled, selected, numerically resolved, or scientifically qualified.

Object	Owns	Must not do
RTC application context	Requested use, exact paired input and interval, named consumers, raw-coordinate convention, and external constraints	Use a context label to enable, disable, or numerically define an operation
RTC resolved plan	Immutable selected operation sequence and exact numerical policy after every required evidence and compatibility predicate passes	Inherit hidden defaults or mutate during apply
RTC realized record	Exact intervals, operations, state, exceptions, outputs, and completion that actually occurred	Rewrite the application context or resolved plan

RTC produces one consumer-neutral atomic bundle. Each downstream consumer binds the members and separately requested diagnostic detail it needs; the bundle schema and raw signal meaning do not change with the consumer label.

2 The learn–resolve–apply paradigm

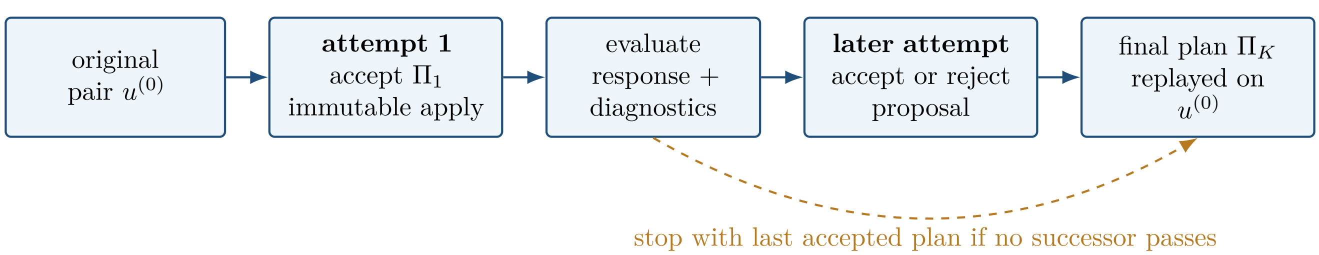
RTC begins with a scientific question: which contaminant or sampling problem must be addressed while retaining adequate astronomical response? The answer is not an operation name. It is a lifecycle that identifies evidence, converts it into a fully specified plan, and applies that plan without hidden adaptation.

Learn. Learn uses a declared population: a separate training interval, a designated part of the same observation, a collection of scans, an observation aggregate, or another named population. It may estimate spike statistics, line centers and widths, filter bands, tap candidates, masks, scan classes, decimation candidates, donor groupings, quality, and uncertainty. For paired streams it may also estimate optical r/x leakage, atmospheric templates, level-shift events, plateau state, and cross-coordinate covariance. Each diagnostic carries estimator, unit, support, selection, and failure state. Learning from the observation later conditioned makes the operator data-dependent. Conditional linear response at fixed state does not eliminate selection bias or parameter-estimation uncertainty.

Resolve. Resolve combines evidence with owner-approved policy. It accepts or rejects every scientific predicate and produces one immutable plan containing sequence, coefficient identity, rates and conventions, order, bands and tolerances, notch state, applicability, masks, donor policy, edges, guards, resets, decimation phase, output timing, expected response/support, uncertainty, and downstream compatibility. Missing evidence or tolerance leaves the dependent plan unresolved.

Apply. Apply consumes exactly one resolved plan. It does not move a notch, change a cutoff or tap count, switch donor policy, change decimation, silently fall back, or refit a parameter. Restart restores the same plan and rejects incompatible inputs. Realized state records what ran without rewriting the request or resolved plan.

V0.1 also permits **bounded iterative notch-plan refinement**. This is an outer finite sequence of actual attempts over complete accepted plans, with a separately configured finite maximum. Parameters may change only after one apply has finished and a successor attempt has admitted a new complete plan. A rejected or stop attempt does not advance the accepted-plan index or manufacture an evaluation product. Each apply remains immutable. This is the v0.1 scientific meaning when "adaptive notch" is used; it is not continuous coefficient tracking and is not the online-adaptive estimator of SCI-RTC-DEF-025.



Attempt a learns from an exact population and the evaluation product of the current accepted plan Π_{k_a}. Resolve returns a complete cumulative proposal $\widetilde{\Pi}_a$ and a typed accepted, rejected, or stop disposition. Only acceptance advances the plan index and creates a new evaluation product. The initial evaluation is explicitly the paired object $u^{\mathrm{eval},(0)} = \mathcal{A}_{\Pi_0}(u^{(0)})$, including when Π_0 is a nonidentity fixed plan. A successor proposal names all retained, modified, removed, or rejected notches and every broad-band, state, mask, edge, timing, and decimation choice. It is never merely "add this notch to whatever ran before." Every accepted evaluation and the final production apply replay the complete pair plan against the original admitted pair $u^{(0)}$. Product projection then publishes required conditioned x and conditioned r only when requested and available. This avoids unintended repeated rounding, padding, state transients, duplicate filtering, stale masks, or ambiguous parentage.

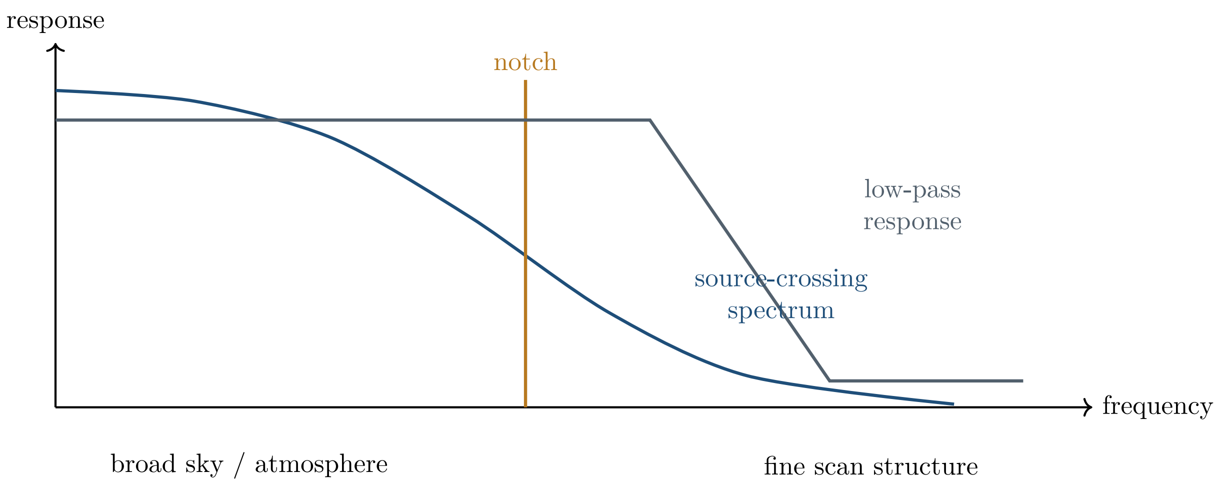
A sequential cascade from the preceding filtered product is a different scientific operator. It is allowed only when selected explicitly and when every immutable intermediate parent, response, state, phase, support, covariance, and uncertainty is retained. Learned-then-fixed and online-adaptive filtering remain different: in the latter, parameters evolve inside one apply pass. V0.1 does not authorize that branch.

Lifecycle test. For each attempt identify its attempt index, current accepted-plan index, learning and parent products, complete proposal, predicates, and disposition. For each accepted plan show that apply did not modify it. Then identify the finite stop cause and the final selected plan. Actual attempts, configured maximum, and accepted plans are different counts; unused maximum attempts are not no-ops. If any link is absent, the refinement claim is unavailable.

3 Temporal frequencies, scan speed, and atmospheric variability

The same temporal frequency represents different angular structure in different scans. For a locally Gaussian effective beam with covariance Σ and angular scan velocity $\mathbf{v}$, the characteristic crossing time is formal identity `SCI-RTC-EQ-023`. It uses the beam projected along the actual scan direction, not merely a scalar array FWHM. Source extent, detector and telescope response, prior RTC stages, and cadence further shape the exact source-times-response spectrum.

High frequencies correspond to finer scan-direction structure. Low frequencies contain broad sky structure together with baselines, atmosphere, and slow instrumental behavior. These do not cleanly separate source and contaminant. A Gaussian is not band-limited; extended emission can share low frequencies with atmosphere; a coherent line can overlap a source crossing. The shorthand $f \sim v/\theta$ is pedagogical only.



Changing speed moves astronomical modes relative to a fixed cutoff or notch; changing direction changes projected beam width. Every filter is assessed over declared speed, direction, beam, source extent, and source class using the exact realized response. Preserved dimensions or units do not establish preserved peak, PSF, centroid, extended response, or absolute mode.

Atmospheric fluctuations are also structured in time, detector, and readout coordinate. A raw atmospheric template may be scientifically useful because it can expose common optical response in x and r, but it is not the sample-dependent target-atmosphere correction owned by SCI-CAL. Its learning population, estimator, frequency support, detector grouping, masks, quality, uncertainty, and applicability all belong to the RTC diagnostic plan.

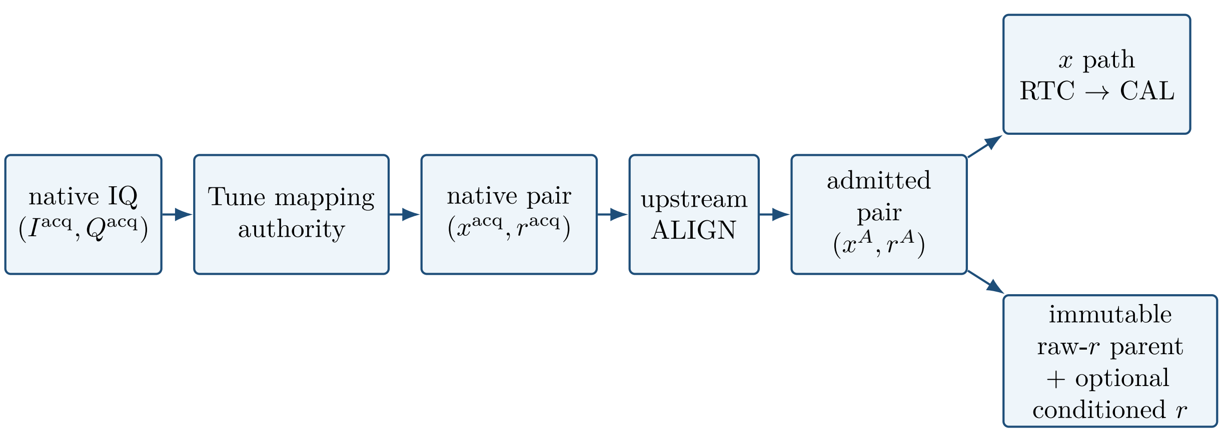
In v0.1 the template is optical excitation and diagnostic evidence only. It may inform leakage, Tune/readout health, artifact and level-shift assessment, or a declared selector, but it is not subtracted from science x. Numerical common-mode removal is a detector-mixing, response-changing estimator with null modes, covariance, source-mask dependence, and astronomical transfer of its own; it requires separate PTC or successor authority. This boundary avoids granting sky cleaning merely because the same atmospheric fluctuations are scientifically useful for measuring leakage.

4 Paired IQ-to-x/r readout coordinates

The acquired scientific object is an ordered pair, not two streams that may later be joined by row number. Tune mapping authority converts measured native in-phase and quadrature coordinates into x_{dj}^{acq} and r_{dj}^{acq} with one detector, sample, stream, and mapping-revision identity. RTC consumes that general mapping and, where applicable, its local Jacobian or affine representation, domain, applicability, validity, and provenance; it does not infer, globally linearize, repair, or silently replace it. Exactly one separate upstream ALIGN relation then produces the admitted x^A/r^A pair. RTC starts there: it retains ALIGN identity and may compose an optional end-to-end response to native acquisition coordinates, but its local operator does not apply ALIGN a second time.

Both members are required for every admitted occurrence. A missing partner is malformed input, but numerical or scientific validity is represented independently: valid r does not repair invalid x, and invalid r does not erase an otherwise valid x. Exact pair identity survives selection, segmentation, coordinate-diagonal filtering/sampling, bundle construction, serialization, and restart without implying that r receives an x-specific repair. Pair-level pathology and affected support coexist with coordinate-specific evidence, numerical availability, and correction amplitudes. The notation m_{native} denotes a native table row so that the physical r coordinate is never confused with a row index.

The output roles differ without making r a subordinate sensor. Raw x is the ordinary science-conditioning and downstream CAL path, but it is not itself Stokes I. Raw r is an equal contamination sensor that can reveal optical response, level shifts, electronics behavior, and shared selection. Expected optical or atmospheric response in r is not automatically contamination: source protection, leakage, Tune, loading, mapping, and causal class all participate before evidence becomes hard pair action. When requested, conditioned r is optional content under the same RTC context, resolved plan, realized record, and exact output grid as conditioned x; it is not an independently optimized pipeline. It keeps raw-coordinate units, sign, reference, mapping, response, support, validity, uncertainty status, leakage lineage, and provenance. V0.1 does not authorize numerical mixing, calibration of r, cross-coordinate donors, or substitution of one coordinate for the other. Polarimetry remains outside scope.



Learn keeps direct-x, direct-r, and genuinely joint evidence separate. Resolve, rather than Apply, admits hard causal classes and unions their action support into one immutable pair plan. The paired consequence is conservative but not causally anonymous: a direct r event may infer pair contamination in x without becoming a direct x detection. Diagnostic and candidate evidence—including weak lines, PSD shape, variance, leakage, coherence, population outliers, Tune health, or source-overlapping features—does not enter the hard union until its stronger selected predicate passes.

Every ordinary canonical stage then uses one identical temporal and sampling operator on both coordinates: in prose, the paired response is the identity in coordinate space tensored with one resolved operator. This identity includes masks, segments, transitions, guards, coefficients, state and reset, filter definition, precision, phase, boundaries, downsampling, selected times, representative occurrences, cardinality, and grid. Both numerical cross-coordinate branches remain zero. A future r-optimized diagnostic would require a separate product name and authority.

The pair has four distinct support concepts. G_{pair} is the common index grid. A_{pair} is the common pair-action and operator support. A_x and A_r are coordinate-local numerical-availability sets and may differ without changing the grid. Joint covariance is asserted only on its declared support $A_C \subseteq A_x \cap A_r$. Paired statistics retain the full x/r covariance on A_C and any shared selector. At fixed resolved state, the paired numerical Jacobian has two identical diagonal blocks and zero cross branches, but available conditioned-pair covariance retains the propagated xr/rx blocks. Data may still follow the causal selection path from r evidence through the pair plan to conditioned x; that dependence is not a numerical $r \to x$ filter branch. Conditional covariance, coordinate-specific offset uncertainty, learned-parameter and selection uncertainty, and total derivative availability remain distinct. At a selector boundary a total derivative may be unavailable. A block-diagonal or independent model requires evidence; it is not implied by coordinate names. This paired boundary applies equally across all RTC application contexts.

5 Despiking, level shifts, and donor replacement

Despikes and level shifts are different contaminants. A spike is localized; a level shift is a persistent paired offset change with a transition, pre/post plateaus, and a new coherent segment. Confusing them can replace real data, bias a plateau, or unnecessarily reset a long filter.

The selected early architecture separates evidence exclusion from numerical replacement. Change-point learning uses the original aligned pair while candidate isolated spikes are excluded, masked, or robustly downweighted. It does not consume donor-substituted values. After level-shift boundaries are resolved, any admitted donor replacement operates only within a stable segment and no donor relation crosses an unresolved or accepted boundary. When despiking is selected, accepted target x cells are actually replaced or recovered under the resolved policy. Hard pathology from either coordinate or genuinely joint evidence produces one cause-preserving pair action support; the evidence origin remains direct or inferred rather than becoming a generic flag. An x repair does not require numerical r repair: the existing donor route is an explicit coordinate-specific exception. It creates no r donor value, and affected conditioned-r locations and response remain present but unavailable over the repair's full transitive influence rather than only its representative cell. Detection without modification is a separate evidence mode, not despiking. The plan records spike statistic, threshold, support, mask/downweighting, false-positive control, shift estimator, donor policy, and resulting boundaries.

A level shift is modeled in v0.1 as an additive baseline change, not as a gain or responsivity change. On each stable plateau, each coordinate is its intended raw-coordinate signal plus its own baseline. One accepted pair event owns the transition, guards, segmentation, and reset, but x and r retain separately estimated amplitudes, uncertainties, quality states, and correction availability. Equality, copying, rotation, or regression of one amplitude from the other is not assumed or authorized. There is no present observational basis for an additional multiplicative degree of freedom. Evidence of a response change therefore blocks the additive interpretation; it does not cause RTC to fit a gain-changing model. Such models remain successor scope unless evidence requires this assumption to be reopened.

The physical/readout event has finite transition width. Each accepted event therefore records a network identity and detector-specific physical-time transition interval, or a conservative bound on that interval, plus timing coherence, separate x and r evidence or typed unavailability, quality, uncertainty, and cause. The corresponding sample set comes from the applicable timing vector. It is never defined as a fixed number of samples, a fraction of a scan or scan leg, or another scan-dependent quantity, so the same physical criterion remains meaningful across cadences and scan strategies.

Transition samples belong to neither stable plateau. They are explicitly flagged, excluded from both plateau estimates, and remain unmodeled even when the stable plateaus are offset-corrected. Later filters, donor replacement, or downsampling may expand their influence according to operator support, but the record preserves the distinction between underlying physical transition support and downstream propagated influence. The transition also delimits coherent segments. Reset is ordinary; carry is a separately authorized continuity exception. Multiple events retain ordered identities under the selected conflict and minimum-segment policy.

"Stable plateau" does not mean a numerically constant timestream. It means consistency with a declared local signal-plus-baseline model accounting for atmosphere, astronomical signal, noise, allowed slow variation, Tune state, valid support, and unresolved changes. Suitability depends on the resolved application context: an interval may support a readout-health diagnostic while being too short for PSD or notch learning, Beammap source inference, or Science. Sample count alone is not authority.

Pre- and post-shift stable plateaus compare their optical-leakage or more general paired response state on declared joint comparison support, or record that comparison as unavailable when optical excitation is inadequate. A material response change rules out an additive-only correction. When both plateaus have scientifically adequate stable support, RTC may estimate a coordinate's additive level difference and translate one plateau to the declared reference. The resolved plan supplies each coordinate's estimator, minimum support, quality and uncertainty criteria, reference plateau, and failure behavior. Each inferred offset applies only to valid stable plateau cells, never to transition support. If one coordinate lacks adequate support, RTC invents no offset for it: the common event remains a segment boundary, its requested conditioned coordinate is unavailable wherever that missing correction can influence a downstream result, and the other coordinate's valid correction and output need not fail. That influence includes the full FIR footprint, IIR state to the next authorized reset, and the support of subsequent sampling or selection; it is not confined to the plateau cell at which correction was needed. If a conditioned-r affine correction is applied, the conditioned record retains it separately while the immutable raw-r parent and original raw-r event diagnostic remain unchanged.

Normal production retains compact spike/shift treatment state and useful population counts and characteristics, not a permanent manifest of every spike, donor link, or fit. Event fits, donor traces, and detailed diagnostics may be verbose products. This changes neither conditioning nor the causes, transition flags, and segment state carried by ordinary products.

A **donor** constructs a continuity surrogate for a rejected target x cell. The target retains identity and replacement cause while donor influence and covariance remain traceable. Under the approved raw convention, compatible `flxscale` values permit the factor ϕ_q/ϕ_d; this transfers convention, not sky identity or calibration. Neither paired coordinate is a donor for the other. Beammap donor factors come from identified prior authority, never the new factor derived from the same Beammap lineage.

For phase-zero output, (d, Mn) on the raw aligned pair is representative. If synthesized or replaced, it is not independent; donors, coefficients, and centroids do not change it; other causes route to consumer policy.

6 Optical leakage diagnostics from atmosphere and bright sources

The r coordinate is not presumed optically dark. A source-dependent local model permits both x and r to respond, and `SCI-RTC-EQ-032` defines a diagnostic ratio only where the x response is valid and nonzero. That ratio is not a universal detector correction. It is an evidence product with a source class, estimator, support, selection, uncertainty, validity, and response convention. Because the number depends on coordinate normalization, every product also retains x/r units or scales, mapping revision, normalization, metric, Tune/loading/epoch domain, and whether the ratio is dimensionless or carries a coordinate-unit ratio. A response angle is comparable only after a declared metric makes the axes commensurate.

Atmospheric variability provides one high-occupancy optical diagnostic. Its estimator must distinguish the selected atmospheric template or population, temporal and frequency support, detector grouping, masks, common-mode model, quality, and uncertainty. A measured x/r correlation is not by itself an unbiased optical-leakage coefficient: the estimator must address shared-data dependence, noise in both coordinates, and detector self-inclusion. Direct r-on-noisy-x regression is generally biased unless the selected method accounts for those effects. Leave-one-detector-out, joint latent-mode, cross-spectral, and other methods remain candidates rather than defaults. Because the atmosphere can share modes with sky signal and electronics, coherence alone does not establish optical origin.

A bright compact source provides a second diagnostic with different spatial and temporal support. It binds the source model, scan geometry, projected beam, background treatment, masks, fit, uncertainty, and source-class applicability. An atmosphere-derived and bright-source-derived value keep separate parentage. Agreement is useful evidence; disagreement is also a result. They are not averaged, substituted, or promoted as mutually validating without an explicit comparison model.

Each diagnostic states whether it is scalar, band-limited, or frequency dependent. A scalar fitted over one band cannot silently become authority outside it. If coordinate-specific preprocessing or filtering changes the two response domains differently, pre- and post-operation leakage estimates are distinct products. Their comparison includes declared joint comparison support and the applicable response. Raw-pair and conditioned-pair diagnostics therefore retain distinct stage identity, mapping, units, response, support, validity, uncertainty, and source class. Ordinary resolved linear filtering propagates nonzero r optical response; it does not rotate or renormalize r toward an ideal dark coordinate. The atmospheric template itself is not subtracted from science x.

A leakage ratio or angle is not compared across detectors, Tunes, loading states, epochs, or mapping revisions merely because both values are finite. The comparison is available only when their coordinate conventions are compatible or an explicit transformation, normalization, and metric have been applied. Thus a change in readout-coordinate scale cannot masquerade as a Tune-health trend.

Expected optical response in r is neither an artifact nor an invalidity. Source protection for artifact learning and correction is pair-coherent. If the required source authority is absent, the affected learned or corrective operation is unavailable; an already resolved applicable linear filter may still act through the identical paired operator. No leakage diagnostic authorizes automatic cross-coordinate correction, calibration of r, or a new donor route. RTC reports the diagnostic and protection facts while PTC and SCI-VAL retain their own decision authority.

7 Notch and broad-band filtering

A notch suppresses a narrowband coherent disturbance while removing the same temporal modes from any astronomical signal. A resolved notch declares center, frequency/rate convention, width or Q, depth, family/order, poles/zeros where applicable, phase, group delay, state, transient, guard, applicability, uncertainty, and drift tolerance. A disappearing line is not by itself evidence that astronomical transfer, centroid, PSF, covariance, or edge support remains adequate.

An r-origin spectral feature is first candidate evidence, not automatic subtraction from either coordinate. Its promotion to a shared canonical notch requires the selected cross-coordinate causal predicate and the complete x astronomical-transfer, phase, support, covariance, source-protection, and false-detection budgets. Once admitted, one identical notch operator acts on both coordinates. Thus RTC unions hard pathology support and may combine spectral candidate evidence, but it does not automatically union spectral subtraction.

Bounded iterative refinement is an outer succession of complete immutable plans. Each actual attempt learns from the paired evaluation of the current accepted plan, itself replayed from original admitted pair $u^{(0)}$, proposes a complete cumulative successor, and records an accepted, rejected, or stop disposition. Acceptance alone advances the plan index. The final plan replays once against $u^{(0)}$ before conditional product projection. A configured maximum is only a bound: an early stop creates no fabricated no-op attempts. Maximum-attempt, oscillating, or parameter-unstable termination is not convergence.

A low-pass may reduce high-frequency noise or prepare for decimation but can attenuate fast source crossings, broaden the PSF, and increase covariance. A high-pass can suppress drift or broad atmosphere while removing extended sky, beam wings, baselines, and absolute modes and producing negative lobes. A band-pass inherits both boundaries. The combined exact response, not the class label or preserved unit, defines what remains.

Learn identifies the contaminant and candidate design over a declared population. Resolve freezes every band, ripple, attenuation, family, order, precision, phase, state, mask, edge, and applicability field. Apply does not retune. Validation varies interference parameters, speed, projected beam, source class and extent, measuring residual, amplitude, centroid, PSF, extended transfer, ringing, support, covariance, and usable fraction.

8 Filter order, phase, registration, and composition

FIR tap count is a scientific design result, not an arbitrary integer. Formal identity `SCI-RTC-EQ-025` chooses the smallest admitted order that passes transfer, alias, phase, support, guard, precision, and realizability predicates. An approximate family formula can initialize a search but cannot replace exact coefficients and response.

More taps can sharpen a transition while increasing support, covariance footprint, edge loss, ringing, and the range affected by a replacement or level-shift guard. Quantization can change ripple, attenuation, notch placement, and stability. Forward/backward filtering changes magnitude, effective order, state, and edge behavior even when ideal interior phase is zero.

Delay creates scan-direction displacement through formal identity `SCI-RTC-EQ-027`. A phase-zero selected grid point does not imply that the preceding response is centered there. Every resolved application context selects zero phase, shifted time, coordinate correction, response-aware downstream use, or a quantitative bound on residual centroid and PSF effect.

Composition order is equally scientific. Spike-candidate masking or robust downweighting informs level-shift learning on the original pair; admitted shifts define state and donor boundaries; any x replacement then occurs inside stable segments; and all ordinary paired RTC operations precede SCI-CAL. Atmospheric-template diagnostics may affect declared selection and uncertainty but are not a numerical removal stage. Each realized x response and, when requested and available, conditioned-r response are composed from the same admitted paired prefilter and selected order. Admission remains governed by the x-domain astronomical transfer and alias budgets.

Validation varies order/taps, family, precision, bands, shift boundaries, diagnostic-template strength, scan speed, beam, and source class while measuring amplitude, shape, delay, ringing, covariance, registration, and usable support. Coefficient algebra alone is not science qualification.

9 Decimation and alias control

V0.1 phase-zero downsampling is paired point selection at slot Mn, not averaging and not anti-alias filtering. When conditioned r is requested, it has exactly the conditioned-x cardinality, selected times, phase, and representative occurrences. Invalid r locations remain in place rather than being dropped, reindexed, interpolated, or filled. Local or global conditioned-r unavailability never damages otherwise valid conditioned x. Sampling changes cadence, Nyquist frequency, temporal identity, support, covariance, aliasing, and coordinate registration. Exact raw r parentage remains in the bundle. The preceding composed filter determines the rejected-band power that remains to fold.

Formal identity `SCI-RTC-EQ-014` includes every folded image. Before resolving M, the exact common paired prefilter must meet approved x-domain transfer and alias bounds over the declared source domain. A Gaussian is not band-limited, so safe means bounded residual transfer, not perfect preservation. Factor choice uses cadence, smallest projected crossing time, maximum valid scan speed, source class, output requirements, timing accuracy, and downstream compatibility.

The representative source occurrence is exactly (d, Mn) on the raw aligned paired parent. That identity is distinct from the assigned output time, the full transitive support, and the largest response coefficient. It drives the direct synthesized/replaced exclusion without converting every nonrepresentative influenced sample into universal ineligibility.

Selected time t_{Mn} does not erase earlier delay or broad support. Tests include paired impulses and tones, on/off-bin frequencies, varied scan speeds/directions, smallest-beam sources, incomplete strides, level-shift boundaries, reset behavior, and exact alias response.

10 Masks, segment boundaries, state, and non-finite handling

A mask is operator control, not acquisition validity or confidence. Source, spike, level-transition, guard, coordinate, and learning masks retain distinct causes, geometry, support, dilation, and boundary conventions. Invalid or wrong-frame coordinates do not become outside-mask samples. A flag does not automatically become a mask, zero weight, or universal eligibility decision.

Gaps, resets, and selected level shifts delimit coherent segments. FIR extension, rejection, reflection, zero padding, and local renormalization are different operators. IIR state requires explicit initialization, reset or carry, transient, stability, and terminal state. Reset is ordinary at an accepted shift. Carry requires separately authorized continuity, complete response, and a residual-contamination criterion; state never crosses merely because a policy field says “carry.” A guard removes or types outputs but does not prove a residual tail negligible without a quantitative criterion.

The realized event and segment state includes ordered shift identities, transition intervals, paired amplitudes, masks, guards, plateau diagnostics, short-segment disposition, ordinary reset, and any separately authorized carry decision. Every operation class remains admitted across application contexts, but only the operations selected and numerically resolved in the immutable plan execute.

A non-finite sample, coordinate, mapping value, factor, coefficient, or state is rejected with a typed cause, represented unavailable with that cause over its exact influence, or handled by an authorized prior recovery rule. It never silently becomes zero or an undifferentiated generic missing value. Tests place impulses, sources, spikes, shifts, masks, and non-finites at every boundary and compare split/rejoined execution against the declared state semantics.

11 Calibration handoff and downstream interpretation

RTC distinguishes detector-static `flxscale`, an atmospheric leakage template used only as diagnostic evidence, sample-dependent target-atmosphere correction, detector mixing, and temporal filtering. RTC remains raw through the complete plan. A compatible valid `flxscale` ratio may transfer donor convention within raw x replacement, but it is not absolute calibration and never involves r.

Only after the complete RTC bundle exists may SCI-CAL consume the exact conditioned x member and apply absolute `flxscale` and target atmosphere. Raw and requested conditioned r remain in their declared detector-coordinate convention for traceability and diagnostics and receive no SCI-CAL operation. Conditioned r shares ordinary applicable RTC stages, not the x-specific repair chain. This boundary does not assert that raw x is Stokes I.

Target-atmosphere correction at one output time is generally not the inverse of attenuation mixed across a temporal filter’s full support. In compact terms, output-time correction followed by an RTC result is not assumed equal to filtering individually atmosphere-corrected inputs. A CAL handoff therefore needs the complete RTC response over that support or a preregistered owner-approved noncommutation bound; a unit label or one output-time atmosphere value cannot supply the missing equivalence.

Filtering can preserve dimensions while changing source peak, PSF, extended response, phase, centroid, and covariance. Calibration cannot restore removed modes. A Beammap-derived factor belongs to its complete RTC lineage; a target conditioned with a different plan needs explicit response calculation, renormalization, or observational evidence before calibration transfer is scientifically available.

The Science consumer route is: “SCI-CAL handoff and any explicitly authorized raw-domain consumers; subsequent PTC, VAL, MAP, and FLT use follows their own contracts.” Pointing, OOF, Beammap, and diagnostic consumers also bind their exact application context, paired parent, resolved plan, and consumer-neutral bundle. RTC supplies required conditioned x, requested conditioned r or its exact availability state, and the immutable raw-r parent, plus exact mapping, cause, support, response, validity, uncertainty, leakage, source-protection, and provenance facts. It does not select PTC joint-r use, aggregate cross-package validity, or assign consumer eligibility, weighting, mapping, or source-estimation policies.

Beammap circularity guard. New `flxscale` derived from a Beammap cannot be used as donor scaling inside that same Beammap RTC lineage; donor factors come from explicit pre-existing authority.

12 Covariance, support, eligibility, validation, and open decisions

Output dependence expands through paired mapping, alignment, spike and shift selection, within-segment donors, atmospheric diagnostic selectors, taps or state, boundaries, and sampling. Support—not storage shape—governs information flow. Reused donors correlate targets, overlapping filters correlate neighbors, shared x/r selectors create cross-coordinate covariance, and an IIR links every output to its declared state boundary.

At fixed realized state, the conditioned-x conditional moment equations use only the x coordinate, preserving its existing upper-left covariance block. Where both conditioned coordinates are requested and available, the response is coordinate-diagonal and an admitted joint covariance preserves the propagated cross-coordinate blocks. Those blocks also remain scientifically necessary because selection, leakage estimation, shift detection, learned templates, or notch parameters may depend on both members. Their additional consequences enter selection, learned-parameter, model, diagnostic, and cross-term uncertainty. Conditioning does not make those terms zero. Unknown covariance remains unavailable rather than diagonal, white, independent, or zero.

RTC need not claim a maximal total covariance. The simpler binding rule is that every covariance or uncertainty claim explicitly enumerates the components and correlations it includes and excludes. Excluded and unknown items are not zero; a conditional or component-limited claim remains validly qualified, while "total" is reserved for declared complete coverage.

Eligibility differs from computability. V0.1 universally excludes an output whose exact representative occurrence (d, Mn) was synthesized or replaced as an independent detector measurement. Elsewhere in full support, RTC preserves value, response, typed cause, linkage, and influence while each consumer applies its own declared policy. Conditioned x, optional same-grid conditioned r or its exact availability state, and immutable raw-r parentage remain incomplete without exact plan, support, response or typed unavailability, influence, validity, uncertainty status, and provenance.

Algebraic fixtures are necessary but not sufficient. Validation varies the scientific design choices and measures the observables they can change.

Study	Variations	Measurements
Notch	center error, drift, width, depth, Q/order, state, direction, overlap	residual line, sky transfer, ringing, transient, amplitude, centroid, covariance, edge loss
FIR/bands	taps, edges, ripple, attenuation, family, precision, speed, beam, source extent	peak/shape, extended transfer, delay, ringing, alias, covariance, usable fraction
Donor	spikes, invalids, factors, sky mismatch, noise, reuse, sources, mask edges	stabilization, bias, covariance, influence extent, scientific data loss
Decimation	on/off-bin tones, speeds/directions, smallest beam, delay, aliases, strides, resets	retained/aliased transfer, timing, coordinate shift, PSF, support, edges
Paired/leakage	mappings, independent validity, atmosphere/source class, direct/joint evidence, hard/candidate classes, scalar/frequency response	pair fidelity, x/r covariance, leakage bias, cause retention, false pair action, x data loss, unavailable states
Level shift	spike adjacency, network/detector times, amplitudes, multiple events, guards, short segments, reset/authorized carry	event accuracy, false detection, plateau bias, state leakage, response and data loss

Each study preregisters domain, estimator, tolerance and falsifier. Passing coefficient algebra does not establish science qualification. Observational performance, science impact and production readiness are separate layers.

The owner ledger retains numerical or methodological authority for mapping tolerances and validity; leakage and level-shift estimators; plateau support; atmospheric leakage diagnostics; learned populations; filter designs; edges/state; decimation/alias; registration; uncertainty; and bundle form. Resolved architecture includes the paired stream, diagnostic-only atmospheric templates, coordinate-specific and joint evidence, cause-preserving hard support union, spike-aware original-pair shift learning before within-segment replacement, required conditioned x with optional same-grid conditioned r and immutable raw-r lineage, one identical ordinary paired operator, coordinate-specific additive offsets, the explicit x-only donor exception, x-only CAL handoff, representative occurrence, honest local/global r unavailability, and downstream ownership. Online adaptation remains deferred. An open entry blocks only its named operation or claim and never authorizes a default or silent fallback. Cross-package follow-up routes CAL, BEAM, AST, PTC, VAL, MAP and filtering questions without letting RTC assume their answers.

Success criterion. Each operation exposes input authority, resolved choice, realized state, consequences, evidence, and unavailable claims.